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RESEARCH ARTICLE

An Observer-Based Vibration Suppression Method for Backlash-Induced Oscillations in EV Powertrains

MANATO KAWASAKI [1], NOBORU SAKAMOTO [1], (Member, IEEE), AKIRA NAKASHIMA [1], (Member, IEEE), YASUHIKO KOBAYASHI [2], AND YUTARO TANAKA [2]

[1] Graduate School of Science and Technology, Nanzan University, Nagoya 466-8673, Japan

[2] Aisin Corporation, Kariya, Aichi 448-8650, Japan

Corresponding author: Manato Kawasaki (kawamana0508@gmail.com)

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ABSTRACT This study proposes an observer-based vibration suppression framework for electric vehicle (EV) powertrains subject to nonlinear gear backlash. A control-oriented nonlinear two-inertia drivetrain model is constructed by incorporating shaft elasticity, viscous damping, backlash characteristics, sensor quantization, measurement noise, and communication delays to reflect practical vehicle conditions. Based on this model, a state observer is developed to estimate key torsional states, including relative displacement and backlash evolution, during rapid transitions between contact and non-contact modes. Using the estimated states, two vibration suppression control methods are designed: Angular Velocity Damping Control (AVDC) and an enhanced version with Impact Mitigation Control (AVDIMC). The proposed framework targets drivability-relevant low-frequency torsional oscillations (typically below 10–20 Hz) during torque reversals, rather than high-frequency gear mesh vibrations associated with Noise, Vibration, and Harshness (NVH) phenomena. Simulations using a high-fidelity exact backlash simulator demonstrate improved torque transmission and vibration suppression compared with conventional disturbance cancellation and motor-side angular velocity feedback control methods. Vehicle experiments further confirm the practical feasibility under real-time constraints, showing reductions in peak driveshaft torsional torque (approximately 2000 Nm to 1000 Nm) and vibration settling time (approximately 1.41 s to 0.53–0.57 s) during torque reversals. These reductions agree well with the simulation results.

INDEX TERMS Backlash, electric vehicle, estimation, observer-based vibration suppression, powertrain.

I. INTRODUCTION

Vibration suppression has long been a central challenge in mechanical and mechatronic systems. In particular, torsional vibrations arising from rotating mechanisms and torque–transmission systems can lead to performance degradation, noise generation, component wear, and even structural failure. As mechanical systems continue to advance toward higher precision, lighter weight, and faster dynamic response, even small vibrations can no longer be neglected because they significantly affect the stability and reliability of the control performance. Consequently, vibration control has become an essential research topic in traditional mechanical engineering and mechatronic applications such as robotics, precision actuators, and electric powertrains [1], [2], [3].

Conventional Internal Combustion Engine (ICE) vehicles, in automotive field, suffer from periodic excitations caused by combustion, torque fluctuations during gear shifting, and torsional resonances of the drive shafts. These vibrations are typically mitigated using mechanical damping devices, such as torque converters, clutches, and dual-mass flywheels. However, the automotive industry is undergoing rapid electrification owing to the global trend toward carbon neutrality. Battery Electric Vehicle (BEV)s, Hybrid Electric Vehicle

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(HEV)s, Plug-in Hybrid Electric Vehicle (PHEV)s, and Fuel Cell Electric Vehicle (FCEV)s are being deployed worldwide, offering high efficiency, low noise, and fast torque response. Simultaneously, new vibration issues unique to electric drive systems have emerged [4], [5], [6].

In electric powertrains, the motor directly drives the reduction gear instead of the engine, resulting in a mechanically stiff and highly responsive coupling. Although this improves the dynamic performance, it also makes the system extremely sensitive to mechanical imperfections, such as gear backlash. Because the motor torque directly reflects the driver's intention, the powertrain control system must respond to torque requests with high accuracy and minimal delays. Any deviation or oscillation in torque transmission impairs drivability and degrades the perceived vehicle quality. Gear backlash refers to the angular clearance between the mating gear teeth, which introduces a dead zone in the torque–transmission path. When the transmitted torque reverses or changes abruptly, one gear rotates freely until contact is re-established, and the subsequent impact generates torsional oscillations in the shaft system [2], [3], [7], [8], [9]. These impacts cause gear rattle, fatigue damage, and degradation of the ride comfort.

Backlash is often modeled either as a static dead-zone nonlinearity or as a dynamic backlash element (e.g., the Nordin-type model) [3], [7]. In a dead-zone model, the backlash effect is represented by a memoryless input–output map, and the backlash state is not treated as a dynamical variable. In contrast, the Nordin-type dynamic backlash model introduces an internal backlash displacement (or relative motion) that evolves over time and governs contact–separation behavior. In this study, we explicitly incorporate the backlash angle θ_b as a dynamical state, estimate it online using an observer, and design the control law based on the estimated backlash state. This observer-based, state-dependent control framework is a key difference from conventional dead-zone-based approaches.

In this study, however, the objective is not to explicitly reproduce detailed high-frequency gear mesh dynamics, but to suppress dominant low-frequency driveline oscillations that directly affect vehicle drivability and ride comfort under real-time control constraints. Here, drivetrain vibration components may span from several tens of hertz to several kilohertz; in particular, gear mesh vibrations associated with time-varying mesh stiffness, tooth contact compliance, and mesh harmonic excitations typically appear in the higher (kilohertz) portion of this range [10], [11]. These higher-frequency components mainly contribute to acoustic noise, local gear stress, and Noise, Vibration, and Harshness (NVH)-related phenomena, and their direct influence on vehicle-level drivability and longitudinal acceleration perceived by drivers is generally limited. In contrast, the present work focuses on the lower-frequency range within this band, typically below approximately 10–20 Hz, which is widely recognized in vehicle dynamics and drivability studies as the dominant frequency range affecting perceived drivability during torque reversals in electric vehicles, rather than NVH characteristics [12], [13], [14], [15]. Transient vibro-impact phenomena induced by torque reversals and gear backlash have been extensively investigated using high-fidelity experimental setups and detailed gear dynamics models [10], [11]. While these studies provide deep physical insight into gear contact phenomena, the present work focuses on control-oriented modeling and real-time vibration suppression rather than detailed gear mesh dynamics.

In ICE vehicles, such vibrations are often masked by engine noise; however, in electric vehicles, the absence of engine excitation makes the backlash-induced vibrations clearly perceptible. Suppressing these vibrations has therefore become crucial for improving the comfort, reliability, and durability of electric vehicles [6], [12], [13], [16], [17], [18], [19]. The two-inertia model is widely used to analyze these dynamics, in which the motor and load sides are represented by separate inertias connected via an elastic shaft or gear mesh. By incorporating the nonlinear characteristics of backlash, the two-inertia model can accurately reproduce torque–transmission behavior during acceleration, deceleration, and regenerative braking [1], [5], [20], [21], [22].

Although an actual electric vehicle powertrain consists of multiple mechanical components such as gears, differentials, and half-shafts, its dominant torsional vibration behavior relevant to vehicle drivability can be effectively represented using a simplified two-inertia model for control-oriented vibration suppression. In this control-oriented representation, the motor-side components are lumped into a single inertia, while the wheel-side dynamics are represented by another inertia connected through an elastic shaft with backlash. Such two-inertia models have been widely adopted in studies on automotive and electric vehicle powertrains, and their validity has been demonstrated through simulation and experimental investigations [12], [13], [18], [23]. While higher-fidelity multi-inertia or gear-mesh models are essential for detailed gear contact analysis, the two-inertia model provides an appropriate balance between physical fidelity and computational tractability for vibration suppression and real-time control design.

Recent studies have highlighted that the nonlinear and discontinuous characteristics of gear backlash significantly influence the torque–transmission behavior and complicate the control system design. To address these challenges, a wide variety of advanced control approaches have been proposed in the literature. Conventional backlash-estimation and compensation schemes have been investigated in automotive drivelines [9], [17], [23], [24], and several model-based analyses of backlash dynamics and driveline oscillations have been reported [4], [5], [6]. In parallel, detailed gear dynamics studies based on high-fidelity modeling and experimental validation have clarified transient vibro-impact phenomena associated with torque reversals and gear backlash [10], [11]. While these studies provide deep physical insight into gear contact and impact mechanisms, they primarily focus on detailed dynamic analysis rather than real-time control implementation. Control strategies such as switched and hybrid control [16], [25], [26], adaptive and mode-switching vibration-suppression techniques [12], [13], [27], and observer-based compensation methods for regenerative braking [18], [24] have also been developed. Furthermore, in electric vehicle applications, model predictive control and learning-enhanced torque-tracking schemes have been introduced to suppress torsional oscillations under nonlinear backlash [21], [22], [28]. Although these approaches have significantly advanced the understanding and mitigation of backlash-induced vibrations, challenges remain regarding robustness to abrupt mode transitions, sensor limitations, and real-time implementation.

Based on these studies, the remaining challenges in real-time vibration suppression can be summarized as follows. Observer-based and predictive control methods, including disturbance- and state-observer schemes as well as Model Predictive Control (MPC), have also been reported [9], [17], [21], [22], [23], [24], [28]. These methods estimate and compensate for unmeasured disturbances or load torques to achieve high-precision vibration suppression. However, they often require accurate model parameters and substantial computational resources, which limits their real-time implementation in automotive embedded systems, particularly under strict sampling-period and computational constraints. Recently, hybrid strategies combining model-based and data-driven approaches have gained attention, including adaptive and preview-based control for electric vehicles [6], [14], [28].

In addition, fuzzy inference-based and nonlinear compensation control schemes have proven effective for systems with uncertainty. Fuzzy rules flexibly express nonlinear relationships between variables and emulate human-like reasoning to achieve adaptive control behavior [13], [27]. Such fuzzy or nonlinear adaptive control schemes have been successfully applied to automotive drivelines and related torsional systems, achieving a trade-off between vibration suppression and fast torque response [13], [26].

Despite these advancements, important challenges remain in designing vibration suppression controllers that are not only effective in simulations but also feasible for real-time implementation in actual electric vehicles. In particular, control-oriented models with reduced complexity are often required to satisfy strict real-time computational constraints, even though they do not explicitly capture all high-frequency drivetrain dynamics. Furthermore, most proposed controllers have been validated only under limited test conditions, restricting their applicability to broader driving scenarios and environments. Parameter tuning in observer-based or adaptive controllers is often empirical or performed offline, limiting practical real-time implementation [9], [12]. In addition, factors such as sampling-period fluctuations, quantization effects, and computational delays are frequently overlooked in control design, despite their substantial influence on vibration-suppression performance [13], [14], [15], [29].

Considering these issues, this study identified four principal design requirements:

1) Compensation for nonlinear and discontinuous torque transmission caused by gear backlash;
2) Estimation of unmeasured states, such as torsional angle and load torque, using an observer;
3) Adaptability to rapid torque transients characteristic of electric vehicles;
4) Robustness against control-period fluctuation and parameter uncertainty.

To fulfill these requirements, this study constructs a two-inertia drivetrain model incorporating gear backlash, shaft elasticity, and torsional stiffness, and develops an observer-based feedback control scheme to estimate the unmeasured states. The proposed approach aims to achieve both high torque responsiveness and effective vibration suppression. In addition to simulation-based evaluation, the effectiveness of the proposed control scheme is experimentally validated using an actual electric vehicle to demonstrate its practical applicability. Because the proposed observer provides highly accurate state estimation, the load-side angular velocity, which is traditionally avoided because of communication delays, measurement noise, and limited sensor resolution [14], [15], [29], [30], can be reliably utilized for vibration suppression. By compensating for these adverse effects through the observer, the proposed method improves the control performance beyond that of conventional approaches.

Although the proposed control framework involves rapid switching between different control modes within very short time intervals, similar mode-switching strategies have been successfully implemented in practical systems with comparable control targets [8], [12], [13], [19], [31]. Thus, the proposed approach remains feasible for real-world applications.

Furthermore, the proposed approach is not limited to automotive powertrains; it can also be extended to general mechatronic systems, such as robotic actuators, machine tools, and industrial servo drives, where nonlinear elastic couplings and backlash are common [3], [7], [32]. Accordingly, this study contributes to the establishment of a unified framework for nonlinear vibration control that is applicable beyond the automotive domain.

The scientific novelty of this study lies in the integration of a control-oriented nonlinear two-inertia drivetrain model, a backlash-state observer applicable under realistic sensing limitations, and observer-based vibration suppression controllers for EV drivetrains. The objective of this study is to estimate backlash-related drivetrain states under practical sensing conditions and suppress torsional oscillations during torque reversals, which is validated through simulation and real-vehicle experiments.

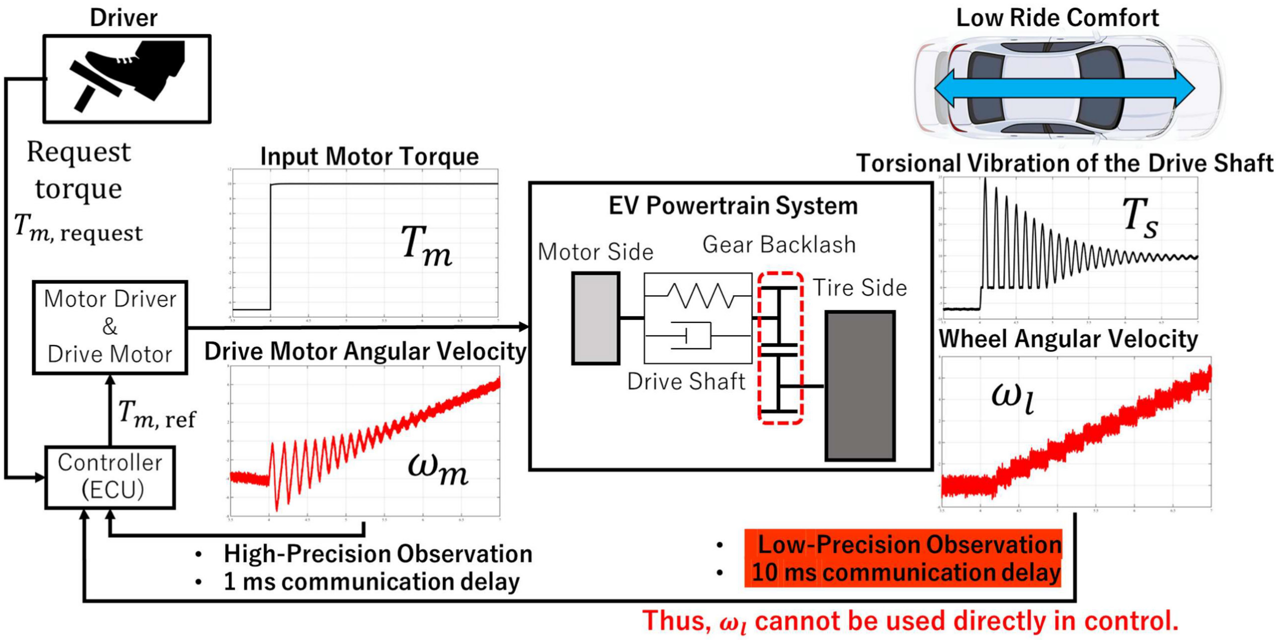

FIGURE 1. Overall control flow in the Electric Vehicle (EV) powertrain considered in this study. The diagram shows the signal path from the driver torque request to motor torque generation, and highlights sensing/communication characteristics (delay and resolution) that motivate the observer-based design.

The remainder of this paper is organized as follows. Section II presents the modeling of the electric vehicle (EV) powertrain, including the representation of the communication delay, measurement noise, and sensor resolution. Section III describes the high-fidelity simulator used in this study. Section IV introduces the observer employed for vibration suppression and explains its formulation and derivation in detail. Section V discusses the conventional and proposed vibration suppression methods for the drive shaft. Section VI provides a simulation-based verification and performance evaluation of the proposed control strategy. Section VII presents experimental validation using an actual electric vehicle. Finally, Section VIII concludes the paper and summarizes its main findings.

II. PROBLEM FORMULATION AND MODEL DESCRIPTION

This section outlines the overall structure of the EV powertrain control system considered in this study. In the target control process, the requested torque of the driver is received by the vehicle control unit (Electronic Control Unit (ECU)), which converts it into the motor output torque delivered to the vehicle. A key requirement of this control system is to accurately interpret the driver's intention and respond to the requested torque rapidly and precisely. Simultaneously, it is essential to suppress the undesired vibrations caused by driveshaft torsion and gear backlash to maintain vehicle handling quality and ride comfort. Accordingly, the control problem addressed in this study is to achieve high responsiveness to the driver's torque request while simultaneously mitigating vibrations originating from mechanical nonlinearities.

Fig. 1 summarizes the overall control flow from the requested torque to the motor torque generation and highlights the characteristics of motor-side and wheel-side measurements, including differences in sensor resolution and communication delay. It also indicates where sensing/communication effects enter the feedback loop, motivating the observer-based design developed in the subsequent sections. Additionally, for the block labeled "Motor Driver & Drive Motor," it should be noted that in actual vehicles, the commanded torque $T_{m,\mathrm{ref}}$ closely matches the generated motor torque T_m. Thus, this block can be regarded as having an approximately unity gain in practical applications.

A. PHYSICAL DYNAMICS AND EQUATIONS OF MOTION

To accurately capture the dynamic behavior of the EV powertrain, the drivetrain was modeled as a two-inertia system with a backlash. In this configuration, the motor-side and load-side angular positions and velocities are coupled through shaft stiffness and damping. The model incorporates the switching nonlinearity associated with the gear backlash, which is essential for reproducing the mechanical behavior observed in real vehicles.

An overview of the two-inertia powertrain model is illustrated in Fig. 2, and the associated variables and parameters are summarized in Tables 1 and 2. The state in which the gear teeth are in contact is referred to as the Contact Mode (CM), whereas the state within the backlash range (free-play region) is referred to as the Backlash Mode (BM). Although an actual vehicle powertrain is a multi-inertia system composed of the motor, reduction gear, differential, driveshafts, hubs, and tires, this study employs a control-oriented equivalent

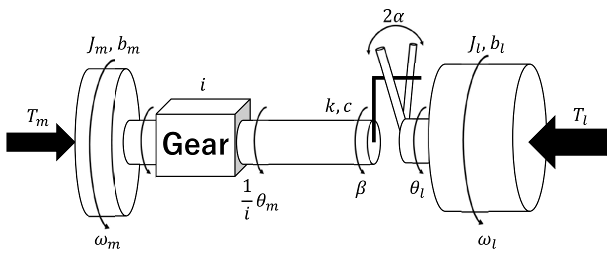

FIGURE 2. Two-inertia powertrain model with backlash. The motor-side and load-side inertias are coupled through the shaft torque T_s, and the backlash element introduces a contact mode (CM) and a free-play mode (BM).

TABLE 1. EV powertrain model variables set.

Symbol	Description	Unit
T_m	Driving motor torque	Nm
T_l	Load torque	Nm
θ_m	Angle on the driving side	rad
θ_l	Angle on the load side	rad
ω_m	Angular velocity on the driving side	rad/s
ω_l	Angular velocity on the load side	rad/s

two-inertia approximation to capture the dominant torsional behavior relevant to drivability. Specifically, the rotating components on the motor and gearbox input side are lumped into the motor-side equivalent inertia J_m, while the drivetrain and wheel-side components downstream of the final reduction are lumped into the load-side equivalent inertia J_l. These two equivalent inertias are coupled by the shaft torsional stiffness and damping (k and c), and the backlash element is modeled as a nonlinear free-play in the torque transmission path, which yields the mode switching between the CM and the BM. To represent the backlash nonlinearity without introducing artificial smoothing, this study adopts the dynamic backlash formulation proposed by Nordin et al. [3], [7], in which the backlash displacement is treated as a state governed by a switching differential equation. This formulation explicitly captures the contact/non-contact transitions and the resulting nonsmooth torque transmission, while remaining suitable for control-oriented analysis and observer-based implementation. This equivalent two-inertia model is not intended to reproduce detailed high-frequency gear-mesh dynamics; rather, it provides a compact representation of the low-frequency driveline torsional oscillations that dominate drivability during torque reversals, and it is therefore suitable for the subsequent observer design and real-time vibration suppression control development.

Therefore, the two-inertia model used in this study should be interpreted as a reduced-order representation suitable for control design and drivability-oriented vibration analysis rather than a complete description of all drivetrain elastic modes.

The equations of motion for the two-inertia system are as follows:

$$J_m\dot{\omega}_m + b_m\omega_m = T_m - \frac{1}{i}T_s, \tag{1}$$

TABLE 2. EV powertrain model parameter set.

Symbol	Description	Value	Unit
J_m	Inertia moment on the driving side	7.0	$\mathrm{kgm^2}$
b_m	Viscous friction coefficient on the driving side	0.0001	$\mathrm{Nm\cdot s/rad}$
J_l	Inertia moment on the load side	400	$\mathrm{kgm^2}$
b_l	Viscous friction coefficient on the load side	0.0005	$\mathrm{Nm\cdot s/rad}$
k	Stiffness coefficient	10000	$\mathrm{Nm/rad}$
c	Damping coefficient	10	$\mathrm{Nm\cdot s/rad}$
α	Backlash angle	0.03	rad
i	Gear ratio	1	-

$$J_l\dot{\omega}_l + b_l\omega_l = T_s - T_l. \tag{2}$$

The variables used are defined in Table 1, and T_s denotes the shaft torsional torque, which is affected by gear backlash. Based on Fig. 2, the relative displacement θ_d and backlash displacement θ_b are defined as

$$\theta_d = \theta_m/i - \theta_l, \quad \theta_b = \beta - \theta_l. \tag{3}$$

Here, β denotes the reference angle associated with the backlash element illustrated in Fig. 2.

Using these, the torsional torque is expressed as

$$T_s = k(\theta_d - \theta_b) + c(\dot{\theta}_d - \dot{\theta}_b) \tag{4}$$

Substituting Eq. (4) into Eqs. (1) and (2), we obtain

$$\dot{\omega}_m = m_1\omega_m + m_2\omega_l + m_3\theta_d + m_4\theta_b + m_5\dot{\theta}_b + m_6T_m, \tag{5}$$
$$\dot{\omega}_l = l_1\omega_m + l_2\omega_l + l_3\theta_d + l_4\theta_b + l_5\dot{\theta}_b + l_6T_l. \tag{6}$$

where

$$m_1 = -\frac{1}{J_m}(b_m + \frac{c}{i^2}), \quad m_2 = \frac{c}{J_m i}, \quad m_3 = -\frac{k}{J_m i}$$
$$m_4 = \frac{k}{J_m i}, \quad m_5 = \frac{c}{J_m i}, \quad m_6 = \frac{1}{J_m}$$
$$l_1 = \frac{c}{J_l i}, \quad l_2 = -\frac{1}{J_l}(b_l + c), \quad l_3 = \frac{k}{J_l}$$
$$l_4 = -\frac{k}{J_l}, \quad l_5 = -\frac{c}{J_l}, \quad l_6 = -\frac{1}{J_l}.$$

Unlike a static dead-zone approximation, the Nordin model describes backlash through state-dependent switching dynamics, which enables the model to reproduce the time evolution of free-play motion and contact re-engagement in a physically consistent manner. This is particularly important for torque reversals in EV drivelines, where repeated contact–separation transitions can generate transient vibro-impact-like torsional responses. According to Nordin et al. [3], [7], the backlash displacement θ_b satisfies the nonlinear switching dynamics as follows:

$$\dot{\theta}_b = \begin{cases} \max\left(0,\ \dot{\theta}_d + \dfrac{k}{c}(\theta_d - \theta_b)\right), & \theta_b = -\alpha \\ \dot{\theta}_d + \dfrac{k}{c}(\theta_d - \theta_b), & |\theta_b| < \alpha \\ \min\left(0,\ \dot{\theta}_d + \dfrac{k}{c}(\theta_d - \theta_b)\right), & \theta_b = \alpha. \end{cases} \tag{7}$$

Eqs. (5) and (6) are represented in a compact form

$$\dot{x}(t) = Ax(t) + Fz(t) + B_1 T_m + B_2 T_l, \tag{8}$$
$$y(t) = Cx(t), \tag{9}$$

where

$$x(t) = \begin{bmatrix} \omega_m & \omega_l \end{bmatrix}^\top, \quad z(t) = \begin{bmatrix} \theta_d & \theta_b & \dot{\theta}_b \end{bmatrix}^\top, \tag{10}$$

and

$$A = \begin{bmatrix} m_1 & m_2 \\ l_1 & l_2 \end{bmatrix}, \quad B_1 = \begin{bmatrix} m_6 \\ 0 \end{bmatrix}, \quad B_2 = \begin{bmatrix} 0 \\ l_6 \end{bmatrix},$$
$$C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad F = \begin{bmatrix} m_3 & m_4 & m_5 \\ l_3 & l_4 & l_5 \end{bmatrix}.$$

The overall EV powertrain model, including the backlash effect, is represented as a nonlinear system formed by Eqs. (7), (8), and (9). The expression in Eq. (10), in which $x(t)$ is unaffected by the backlash while $z(t)$ is, was originally proposed in [24].

B. SENSING, COMMUNICATION, AND DELAY MODELING

In addition to the physical dynamics, this study explicitly models the sensor noise, quantization effects, and communication delays, as illustrated in Fig. 1. These effects play a critical role in real vehicle applications.

The motor-side and wheel-side angular velocities exhibited the following characteristics, with the corresponding numerical values shown in Table 3.

- **Motor-side sensing:** Measured using high-resolution encoders or resolvers, providing low latency and high signal-to-noise ratio [14], [15], [29].
- **Wheel-side sensing:** Hall-effect or variable reluctance wheel speed sensors (Wheel Speed Sensor (WSS)) are typically used, resulting in lower resolution due to the limited number of tone-wheel teeth [14], [30].
- **Communication delays:** Motor-side signals have minimal delay because they are processed internally, whereas WSS signals transmitted over the CAN bus exhibit delays of approximately 10–20 ms [15], [29], [30].

Motor-side measurements are treated as effectively unquantized owing to their inherently high precision. In contrast, wheel-side measurements are quantized to reflect the limited resolution of in-vehicle wheel-speed sensing. While production WSS signals typically exhibit coarse resolution on the order of several rad/s (Table 3), in this study the quantization step size was set to 0.5 rad/s to emulate sensing/communication limitations under the experimental and simulation conditions, as summarized in Table 4. The observation noise was modeled with signal-to-noise ratios of 20 dB (motor side) and 15 dB (wheel side), consistent with Table 4.

III. EXACT BACKLASH SIMULATOR AND ITS VALIDATION

In this study, an Exact Backlash Simulator developed by Takeda and Nishida [20] was employed. This simulator reproduces the physical behavior of gear backlash by numerically solving nonlinear differential equations based on Nordin's model (7), explicitly considering contact and non-contact states of the gear teeth. Compared with simulators that adopt a conventional dead-zone approximation, this simulator enables a more precise evaluation of drivetrain dynamics, which is particularly important when shaft torsion and torque reversals occur in EV applications.

The simulator adopts an event-based numerical integration framework in which the system dynamics are integrated continuously within each contact mode, while mode transitions are detected based on the backlash boundary conditions. Specifically, switching events among all the modes in Eq. (7) are identified by monitoring the relevant quantities such as θ_b and $\dot{\theta}_d + (k/c)(\theta_d - \theta_b)$, and the governing equations are updated accordingly at each detected event. This approach allows accurate handling of nonsmooth dynamics associated with gear impacts without introducing artificial smoothing. Details of the numerical implementation and validation of the simulator are provided in [20].

Furthermore, because both the proposed and conventional control methods are evaluated using the same simulator, observation noise, communication delay, and sensor resolution modeled in Section II-B were configured to closely replicate those of actual vehicles. As a result, dynamic phenomena such as transient torsional vibrations can be reproduced with sufficient fidelity for control performance evaluation. The simulation results are presented in the following sections.

The simulation scenario corresponds to a tip-in maneuver, in which acceleration occurs immediately after deceleration due to regenerative braking. Fig. 3 summarizes representative responses of the exact backlash simulator during this maneuver. Fig. 3(a) shows the motor torque input T_m, which reverses from negative (regenerative braking) to positive (drive torque). Fig. 3(b) shows the motor-side angular velocity ω_m, where oscillatory components appear right after the torque reversal. Fig. 3(c) shows the load-side angular velocity ω_l; its staircase-like appearance reflects the modeled sensor resolution and measurement noise consistent with in-vehicle wheel-speed sensing. Fig. 3(d) shows the resulting driveshaft torsional torque T_s, where transient peaks and oscillations correspond to repeated contact–separation transitions and vibro-impact behavior induced by backlash. Finally, Fig. 3(e) shows the backlash angle θ_b, which is bounded by $\pm\alpha$ and indicates the evolution

TABLE 3. Comparison of observation characteristics in production EVs. The table summarizes typical communication delay, sensor resolution, and noise level for motor-side sensing and WSS-based wheel-side sensing used in production vehicles.

	Communication delay	Resolution (Quantization step size)	Noise (SNR)
Motor-side encoder / resolver	Low delay [14], [15], [29]	High-resolution (sub-rad/s) [14]	High SNR [14]
Wheel-side WSS	10–20 ms [30]	Coarse resolution (several rad/s) [30]	Lower SNR [30]

TABLE 4. Communication delays, sensor resolution, and observation noise considered in this study. The values are used to emulate sensing/communication limitations in vehicle environments for the simulation-based evaluation.

	Communication delay	Resolution (Quantization step size)	Noise (SNR)
Motor-side	1 ms	Without quantization	20 dB
Wheel-side	10 ms	0.5 rad/s	15 dB

of the drivetrain motion within the backlash (free-play) region.

In addition, Fig. 4 shows the driveshaft torsional torque T_s together with the simulator mode transitions. Fig. 4 indicates repeated switching between left- and right-contact modes during the transient phase, which corresponds to gear impacts caused by torque reversals. Such transient vibro-impact behavior induced by backlash and torque reversals has been widely reported in both detailed gear dynamics studies [10], [11] and control-oriented investigations focusing on electric vehicle drivelines [12], [13]. The practical effectiveness of the proposed control approach in real vehicles is further demonstrated through vehicle experiments presented in Section VII.

IV. BACKLASH ESTIMATION OBSERVER

To suppress the vibrations occurring in the system in Eqs. (7), (8), and (9), it is necessary to estimate physical variables ω_m, ω_l, the backlash angle θ_b and its derivative $\dot{\theta}_b$ as well as the load torque T_l. Taking the drive torque T_m as the input u, the system is augmented with

$$\dot{T}_l = 0.$$

A. OBSERVER EQUATIONS

The key step in designing the observer is to handle the dynamic nonlinearity in Eq. (7) while keeping its structure as simple as possible for the actual implementation. The observer states are the estimates of the drive and load angular velocities, and the load torque:

$$\hat{x}(t) = \begin{bmatrix} \hat{\omega}_m & \hat{\omega}_l & \hat{T}_l \end{bmatrix}^\top .$$

Define $\hat{\theta}_m$, $\hat{\theta}_l$, and $\hat{\beta}$ as the estimates of θ_m, θ_l and β, respectively, and let

$$\hat{\theta}_d = \hat{\theta}_m / i - \hat{\theta}_l, \quad \hat{\theta}_b = \hat{\beta} - \hat{\theta}_l \tag{11}$$

denote the estimates of the backlash-related variables corresponding to the second equation for z in Eq. (10).

Our observer takes the following form:

$$\dot{\hat{x}}(t) = \hat{A}\hat{x}(t) + \hat{F}\hat{z}(t) + \hat{B}u(t) + K(y(t) - \hat{y}(t)) \tag{12}$$

$$\hat{y}(t) = \hat{C}\hat{x}(t), \tag{13}$$

where matrices $\hat{A}$, $\hat{B}$, $\hat{C}$, and $\hat{F}$ are defined as

$$\hat{A} = \begin{bmatrix} m_1 & m_2 & 0 \\ l_1 & l_2 & l_6 \\ 0 & 0 & 0 \end{bmatrix}, \; \hat{B} = \begin{bmatrix} m_6 \\ 0 \\ 0 \end{bmatrix}$$

$$\hat{C} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \; \hat{F} = \begin{bmatrix} m_3 & m_4 & m_5 \\ l_3 & l_4 & l_5 \\ 0 & 0 & 0 \end{bmatrix}.$$

Observer gain K is given by $K = P\hat{C}^T R^{-1}$, where P is the stabilizing solution to the following Riccati equation:

$$\hat{A}P + P\hat{A}^T - P\hat{C}^T R^{-1}\hat{C}P + Q = 0,$$

where Q and R are positive definite matrices appropriately chosen for error convergence. We note that the pair $(\hat{C}, \hat{A})$ is observable and is not affected by the backlash nonlinearity. The idea of separating the backlash nonlinearity from the observer states was originally proposed in [24]. The second term $\hat{F}\hat{z}$ in Eq. (12) plays a crucial role in the convergence of the error, which is discussed in detail in the following subsection, and is defined as

$$\hat{z}(t) = \begin{cases} \begin{bmatrix} \hat{\theta}_d(t), & -\alpha, & 0 \end{bmatrix}^\top, & \hat{\theta}_d(t) \leqq -\alpha \\ \begin{bmatrix} \hat{\theta}_d(t), & \hat{\theta}_d(t), & \dot{\hat{\theta}}_d(t) \end{bmatrix}^\top, & |\hat{\theta}_d(t)| < \alpha \\ \begin{bmatrix} \hat{\theta}_d(t), & \alpha, & 0 \end{bmatrix}^\top, & \hat{\theta}_d(t) \geqq \alpha. \end{cases} \tag{14}$$

Eq. (14) represents the core logic of the observer. In the first and third cases (i.e., in the CMs), θ_b and $\dot{\theta}_b$ are replaced by $\pm\alpha$ and 0, respectively. However, in the middle case (i.e., in the BM), $\hat{z}$ is not an estimate of z in Eq. (10), which includes the backlash angle θ_b, instead, the estimates $\hat{\theta}_d$, $\dot{\hat{\theta}}_d$ of θ_d, $\dot{\theta}_d$, respectively, are employed. This avoids using the backlash angle, which is the source of the troublesome nonlinearity, simplifies the observer structure, and makes it implementable

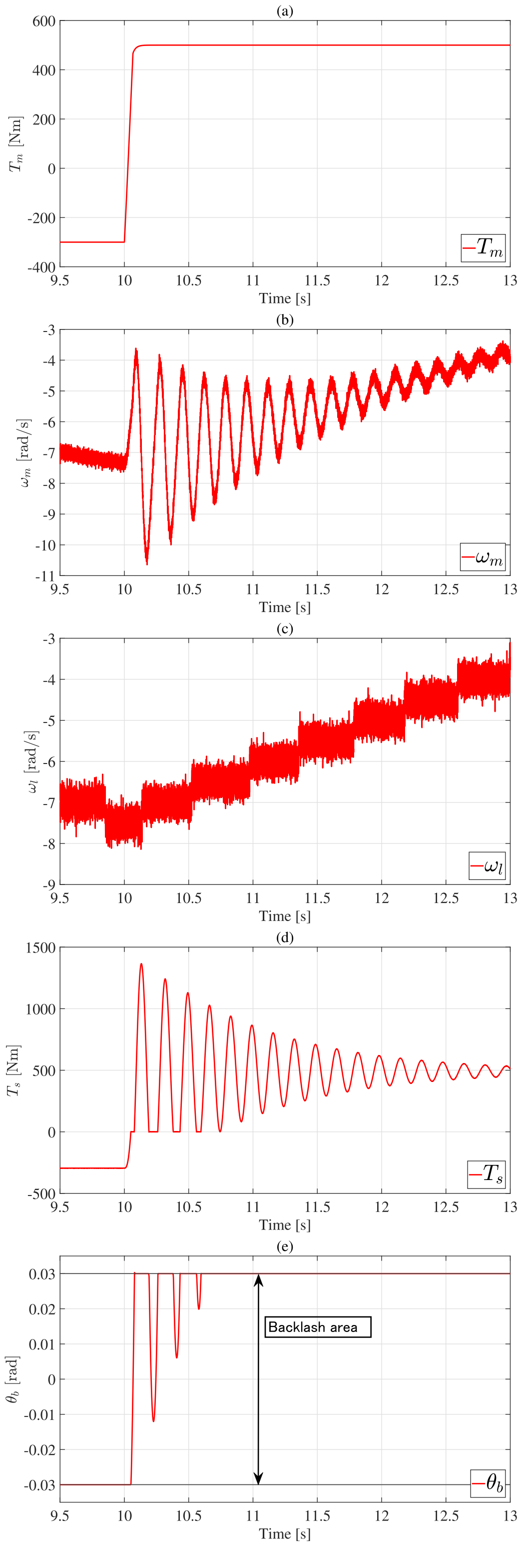

FIGURE 3. Representative responses of the exact backlash simulator during the tip-in maneuver. The plots show the torque reversal input and the resulting drivetrain responses, including angular velocities, shaft torsional torque, and the backlash state evolution.

in real systems. The justification for the replacements $\theta_b \leftrightarrow \theta_d, \dot{\theta}_b \leftrightarrow \dot{\theta}_d$ is based on the physical analysis in Appendix A.

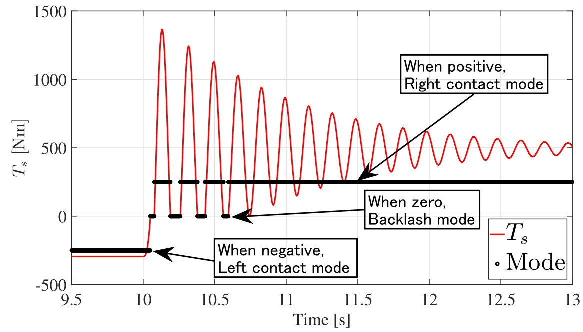

FIGURE 4. Relationship between the driveshaft torsional torque T_s and the simulator modes. The plot shows mode transitions associated with contact and free-play states, which occur repeatedly during the transient after torque reversal.

This approximation is valid when the torsional deformation in the shaft rapidly decays compared with the dominant drivetrain vibration period, allowing the motor-side and load-side angles to be treated as locally equivalent during the backlash mode.

B. CONVERGENCE ANALYSIS OF ESTIMATION ERROR

The estimation error $e = \left[\omega_m - \hat{\omega}_m,\ \omega_l - \hat{\omega}_l,\ T_l - \hat{T}_l\right]^\top$ satisfies

$$\dot{e}(t) = (\hat{A} - K\hat{C})e(t) + \hat{F}(z(t) - \hat{z}(t)), \tag{15}$$

where z is given by Eq. (10). Note first that $z - \hat{z}$ in the last term in Eq. (15) becomes

$$z(t) - \hat{z}(t) = \begin{cases} \left[\theta_d(t) - \hat{\theta}_d(t),\ 0,\ 0\right]^\top, & \theta_d(t) \leqq -\alpha \\ \left[\theta_d(t) - \hat{\theta}_d(t), \theta_b - \hat{\theta}_d(t), \dot{\theta}_b - \dot{\hat{\theta}}_d(t)\right]^\top, & \\ & |\theta_d(t)| < \alpha \\ \left[\theta_d(t) - \hat{\theta}_d(t),\ 0,\ 0\right]^\top, & \theta_d(t) \geqq \alpha. \end{cases}$$

Convergence in the CMs: In these modes, the error equation (15) is

$$\dot{e}(t) = (\hat{A} - K\hat{C})e(t) - \tilde{F}(\theta_d(t) - \hat{\theta}_d(t)), \tag{16}$$

where $\tilde{F} = \left[m_3\ l_3\ 0\right]^\top$. It is seen, from Eqs. (3), (11), that $s(t) := \theta_d(t) - \hat{\theta}_d(t)$ satisfies

$$\dot{s}(t) = He(t), \text{ where } \quad H = \left[\tfrac{1}{i}\ -1\ 0\right].$$

and Eq. (16) is written as $\dot{e}(t) = (\hat{A} - K\hat{C})e(t) + \tilde{F}s(t)$. Now, $w(t) := \left[e(t)^\top\ s(t)\right]^\top$ satisfies

$$\dot{w}(t) = \mathcal{A}w(t), \text{ where } \quad \mathcal{A} = \begin{bmatrix} \hat{A} - K\hat{C} & \tilde{F} \\ H & 0 \end{bmatrix} \in \mathbb{R}^{4\times 4}.$$

The observer gain K was designed such that both $\hat{A} - K\hat{C}$ and $\mathcal{A}$ are stable matrices. The values of K and the eigenvalues of these matrices are

$$K = \begin{bmatrix} 0.258 & 0.282 \\ 0.287 & 0.364 \\ -0.182 & -0.231 \end{bmatrix},$$

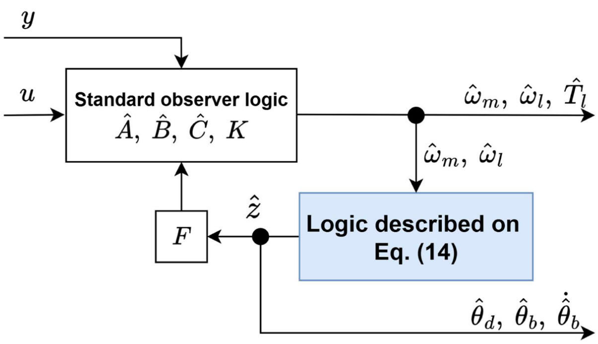

FIGURE 5. **Structure of the proposed observer. The figure summarizes the standard observer part and the switching logic based on $\hat{\theta}_d$ used to generate $\hat{z}$ in Eq. (14).**

$$\lambda_{\hat{A}-K\hat{C}} = \{-2.1331,\ -0.6679,\ -0.0017\},$$
$$\lambda_{\mathcal{A}} = \{-1.0767 \pm 38.1109j,\ -0.6477,\ -0.0016\}.$$

Since the matrix $\mathcal{A}$ is stable, the estimation error $e(t)$ as well as $\theta_d(t) - \hat{\theta}_d(t)$ converge to zero in the CMs.

Analysis during the BM: In this mode, we employ the approximation in Appendix A, where $\theta_b \approx \theta_d$ and $\dot{\theta}_b \approx \dot{\theta}_d$ hold locally. Therefore, the estimation error dynamics in Eq. (15) becomes independent of the backlash θ_b.

Mode switching and convergence: We exploit the oscillatory nature of the system (see Fig. 4). As soon as the mode switches from the BM to the CM, the estimation error converges to zero. Thus, once the system enters the CM, $z(t) - \hat{z}(t)$ remains small, and the estimates $\hat{\theta}_d$, $\hat{\theta}_b$, and $\dot{\hat{\theta}}_b$ are obtained.

C. OBSERVER STRUCTURE

The structure of the observer is shown in Fig. 5. Note that the observer in Eqs. (12) and (13), except for the term $\hat{F}\hat{z}$, has the standard form consisting of $\hat{A}$, $\hat{B}$, $\hat{C}$, and K. The main observer logic in Eq. (14) is further simplified in Fig. 5, where the estimate $\hat{\theta}_d$ is used instead of θ_d in the logic-switching conditions. This simplified observer is used throughout this study.

D. ESTIMATION ACCURACY OF THE OBSERVER BASED ON EXACT SIMULATOR

The simulation assumed a tip-in maneuver, in which the system transitioned from deceleration to acceleration. To emulate real vehicle conditions, the measurements included communication delays, noise, and quantization, as described in Section II-B.

Fig. 6 shows the estimation results of the proposed observer obtained from the exact backlash simulator. In this figure, the red solid lines indicate the observed (measured) signals, the blue dashed lines denote the estimated values obtained by the observer, and the gray lines represent the reference values without noise or quantization. Fig. 6(a) shows the motor torque T_m, which serves as the system input during the tip-in operation. Fig. 6(b) shows the motor-side angular velocity ω_m, where the estimated response closely follows the reference signal despite the presence of transient oscillations immediately after torque reversal. Fig. 6(c) shows

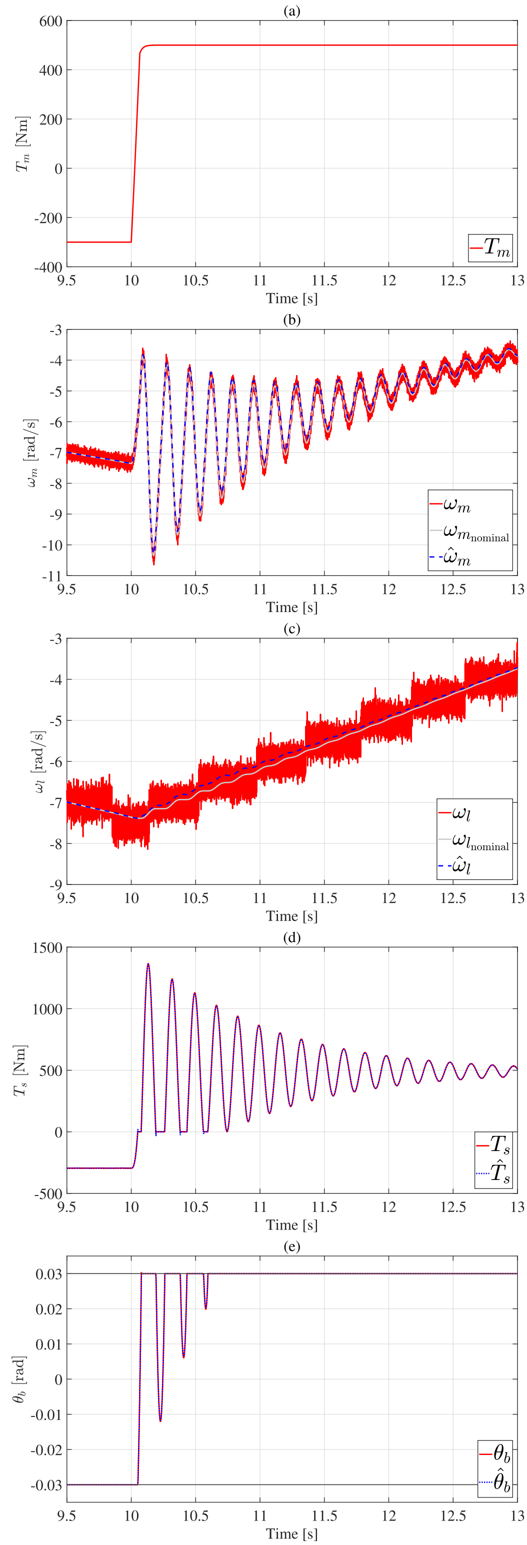

FIGURE 6. **Estimation results of the observer during the tip-in maneuver. The figure compares measured signals (red), observer estimates (blue dashed), and noise-free reference values (gray) for the key drivetrain variables.**

TABLE 5. Root mean squared error (RMSE) between true values and estimated values from the simulation. The errors are computed for the main variables estimated by the observer under the sensing and delay conditions described in Section II-B.

Variables	RMSE	Unit
Motor-side angular velocity ω_m	2.438×10^{-2}	rad/s
Load-side angular velocity ω_l	2.412×10^{-2}	rad/s
Torsional torque of the drive shaft T_s	8.372×10^{-1}	Nm
Backlash angle θ_b	1.027×10^{-5}	rad

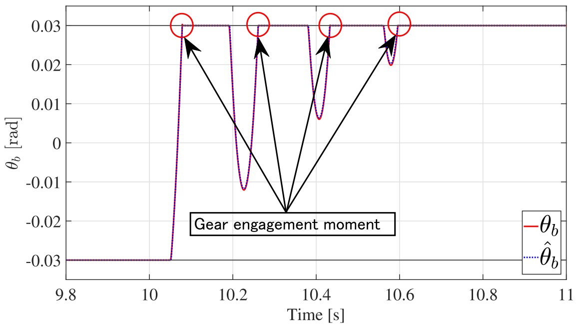

FIGURE 7. Enlarged view of the backlash angle around the gear engagement instant. The figure corresponds to Fig. 6(e) and is used to evaluate the engagement timing accuracy of the observer.

the load-side angular velocity ω_l, in which the discretized appearance originates from the modeled sensor resolution and communication effects; nevertheless, the estimated signal accurately reconstructs the underlying continuous dynamics. Fig. 6(d) shows the driveshaft torsional torque T_s, demonstrating that the observer captures the transient vibro-impact behavior associated with backlash-induced contact and separation. Finally, Fig. 6(e) shows the backlash angle θ_b, confirming that the observer accurately estimates the backlash evolution within the physical bounds $\pm\alpha$.

Table 5 summarizes the root mean squared errors (RMSEs) between the true values and the estimated values for ω_m, ω_l, T_s, and θ_b obtained from the simulation. Together with Fig. 6, the results confirm that all estimated variables closely follow the reference values with sufficiently small estimation errors under realistic sensing and communication conditions.

In addition, Fig. 7 provides an enlarged view of the backlash angle around the gear engagement instant corresponding to Fig. 6(e). This enlarged view shows that the timing of gear engagement is estimated with an average accuracy of approximately 0.4 ms. Considering that the control cycle between ECUs in real vehicles is typically on the order of 1 ms, the achieved estimation accuracy is sufficient for practical real-time applications.

V. VIBRATION SUPPRESSION CONTROL

In this study, we propose a vibration suppression method using the backlash estimation observer introduced in Section IV. To compare the control performance, two conventional control methods commonly used in two-inertia systems are presented. In the next section, a comparison with these methods is conducted using the exact simulator.

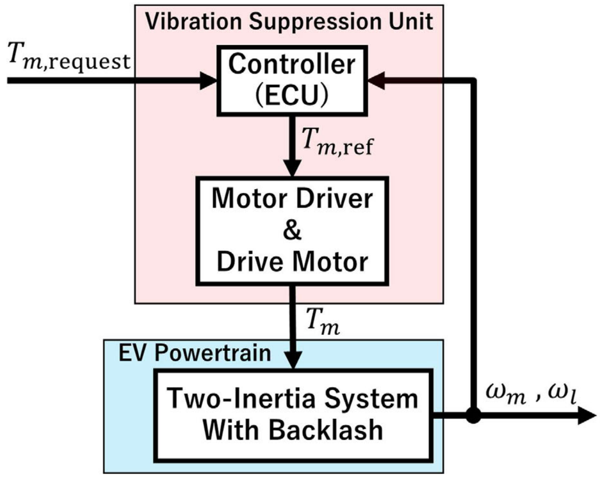

FIGURE 8. Overall system block diagram including the conventional controllers. The figure shows the signal path and the locations of the two conventional feedback loops used for comparison in this study.

A. CONVENTIONAL METHODS

A block diagram of the overall system including the conventional controllers is shown in Fig. 8. Fig. 8 summarizes the signal flow from the torque request to the motor torque generation and indicates where each conventional feedback loop is applied.

Before introducing the proposed method, two conventional vibration suppression control methods are presented for comparison. These methods are feedback control based on the motor angular velocity ω_m, which is easy to design and implement, and feedback compensation-based control for disturbance rejection [19], [31]. The details of these methods are described below.

1) MOTOR-SIDE ANGULAR VELOCITY FEEDBACK CONTROL (ω_M FB)

This method feeds back only the resonance-frequency component of the motor angular velocity ω_m, which is a measurable signal. The controller structure consists of a Band-pass Filter (BPF) centered at the resonance frequency of the drive shaft, as shown in Eq. (17), and a feedback gain D_m.

$$G_{\text{BPF}} = \frac{2(1-\zeta_p)\omega_p s}{s^2 + 2\omega_p s + \omega_p^2} \tag{17}$$

$$\zeta_p = \frac{c}{2}\sqrt{\frac{J}{k}}, \quad \omega_p = \sqrt{Jk}; \quad J = \frac{J_m + J_l}{J_m J_l} \tag{18}$$

The block diagram of the Motor-side angular velocity feedback control (ω_m FB) controller is shown in Fig. 9. Fig. 9 illustrates the BPF and the feedback path used to generate the damping action around the resonance frequency.

TABLE 6. Design parameters and values for the ω_m FB.

Parameter	Value
D_m	200
ζ_p	0.0286
ω_p	38.126

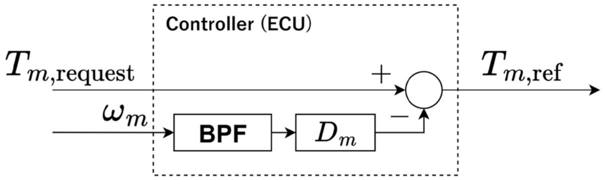

FIGURE 9. Block diagram of the ω_m FB. The controller extracts the resonance component of ω_m using a BPF and feeds it back with gain D_m.

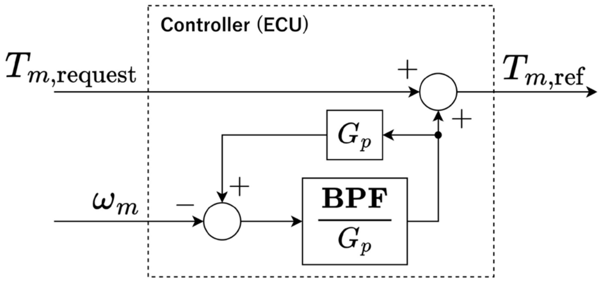

FIGURE 10. Block diagram of The Disturbance cancellation feedback control (DCC). The feedback loop combines the plant dynamics G_p and the BPF to suppress resonance components in ω_m via disturbance cancellation.

2) DISTURBANCE CANCELLATION FEEDBACK CONTROL (DCC)

This method employs a feedback system, as shown in Fig. 10, which consists of the transfer function G_p from the input torque T_m to the motor angular velocity ω_m (based on Eqs. (1) and (2)) and the BPF G_{BPF} in Eq. (17). This control approach is used in hybrid and electric vehicles [19], [31].

When no control is applied, the open-loop frequency response of G_p exhibits a peak around 9 Hz, corresponding to the natural frequency of the system, as shown by the black line in Fig. 11. In contrast, the closed-loop system $\tilde{G}_p$ from T_m to ω_m with the DCC applied suppresses this resonance component, as indicated by the red line.

B. PROPOSED METHODS

From now on, we use the notations $\omega_d = \dot{\theta}_d$, $\omega_b = \dot{\theta}_b$, $\hat{\omega}_d = \dot{\hat{\theta}}_d$, and $\hat{\omega}_b = \dot{\hat{\theta}}_b$. Two observer-based control approaches are proposed in this section, and their configurations are illustrated in Fig. 12. Fig. 12 shows how the backlash observer provides the estimated variables used for feedback and mode switching.

The first is a fundamental control strategy, AVDC, for two-inertia systems, in which the estimate $\hat{\omega}_d$ of the differential angular velocity ω_d is fed back to suppress vibrations. The second is AVDIMC, which utilizes the observer information and aims at a better torque response by minimizing the time duration of the BM. This approach switches the control between the CMs and the BM based on the estimated backlash angle $\hat{\theta}_b$. Furthermore, during the BM, it switches between a fast response control to exit the BM quickly and a virtual damping control that mitigates the shock at gear re-engagement.

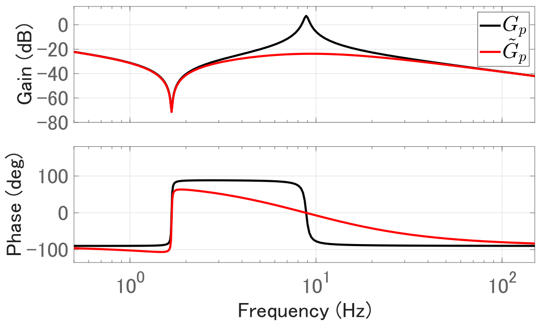

FIGURE 11. Frequency response of G_p with and without the DCC. The figure compares the open-loop response (without control) and the closed-loop response (with the DCC) around the drivetrain resonance.

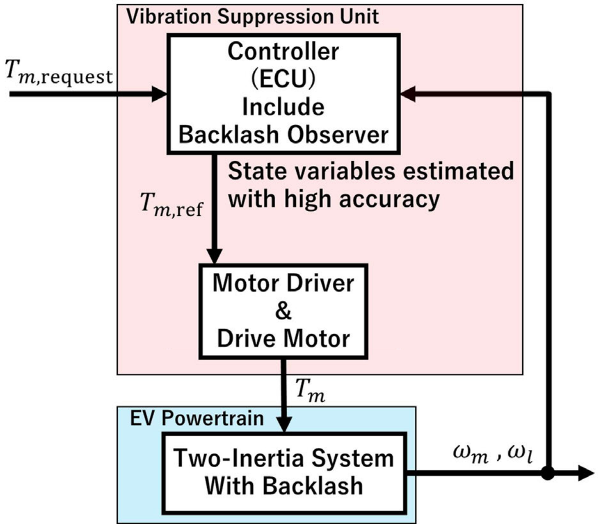

FIGURE 12. Overall system block diagram including the proposed controllers with the backlash observer. The observer outputs are used to compute $\hat{\omega}_d$ for Angular velocity damping control (AVDC) and to determine the mode-dependent control logic for AVD with impact mitigation control (AVDIMC).

1) THE AVDC

This employs a simple negative feedback of the estimated differential angular velocity $\hat{\omega}_d$, which is calculated from the estimated motor angular velocity $\hat{\omega}_m$ and the estimated load-side angular velocity $\hat{\omega}_l$.

The gain D_a is designed as follows. Suppose that ω_d is available and let $T_m = -D_a\omega_d + v$ in Eqs. (1) and (2).

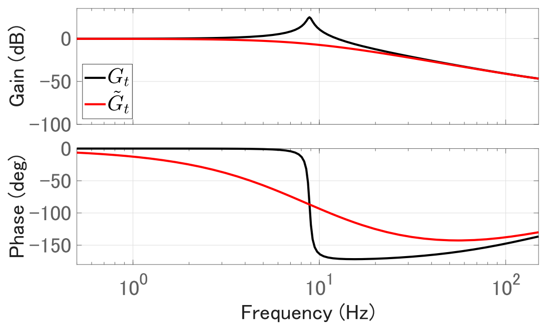

FIGURE 13. Frequency response of the drivetrain with and without ω_d feedback. The figure compares the resonance peak in the baseline response and its suppression achieved by the proposed AVDC feedback.

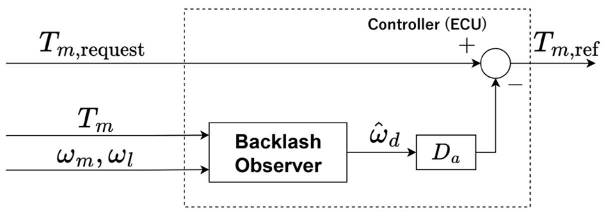

FIGURE 14. Block diagram of AVDC. The controller generates the torque command by adding a damping term proportional to the estimated differential angular velocity $\hat{\omega}_d$.

Then, the closed-loop transfer function from v to T_s is approximately computed, using $b_m = 0.0001 \approx 0$, $b_l = 0.0005 \approx 0$, as

$$\frac{\frac{c}{J_m}s + \frac{k}{J_m}}{s^2 + (Jc + \frac{D_a}{J_m})s + Jk} = \frac{\frac{c}{J_m}s + \frac{k}{J_m}}{s^2 + 2\omega_p\zeta s + \omega_p^2},$$

where $\zeta = \zeta_p + \frac{D_a}{2J_m\omega_p}$ and ω_p, ζ_p, and J are given in Eq. (18). The gain D_a is chosen in such a way that the transfer function does not have a resonance peak (i.e., $\zeta = 1$), and

$$D_a = 2J_m\omega_p(1 - \zeta_p).$$

The frequency-domain characteristics of the proposed angular-velocity damping control are examined through frequency response analysis. Fig. 13 shows the frequency response of the closed-loop transfer function $\tilde{G}_t$ from the auxiliary input v to the driveshaft torsional torque T_s. For comparison, the frequency response of the transfer function from the motor torque T_m to the motor-side angular velocity ω_m, derived from Eqs. (1) and (2) without ω_d feedback, is also plotted.

Using the estimate $\hat{\omega}_d$, the control input $T_{m,\mathrm{ref}}$ of the AVDC is expressed as follows:

$$T_{m,\mathrm{ref}} = T_{m,\mathrm{request}} - D_a\hat{\omega}_d. \tag{19}$$

The block diagram of the AVDC controller is shown in Fig. 14. Fig. 14 illustrates how $\hat{\omega}_d$ from the observer is fed back to generate the damping torque term.

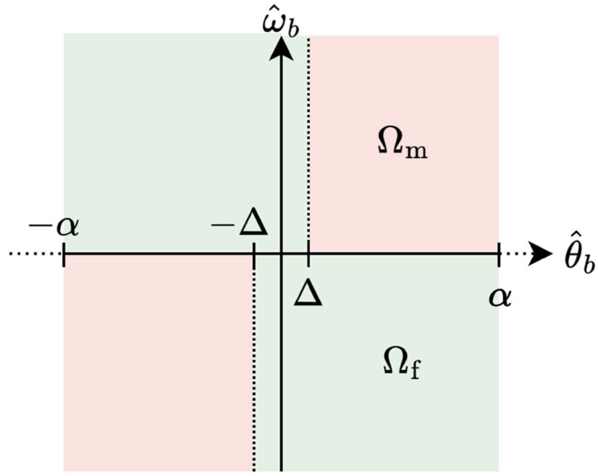

FIGURE 15. Control regions for AVDIMC in the $\hat{\theta}_b$–$\hat{\omega}_b$ plane. The figure defines Omega$_f$ for fast exit from the backlash region and Omega$_m$ for impact mitigation near gear re-engagement.

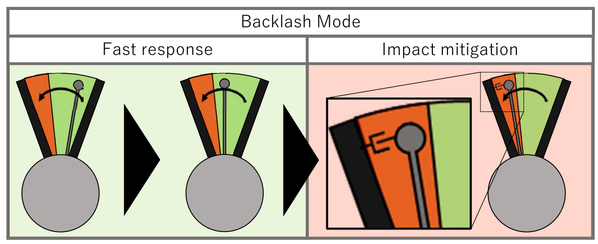

FIGURE 16. Illustration of drivetrain behavior under AVDIMC. The left side corresponds to fast response control in Omega$_f$, and the right side corresponds to impact mitigation control in Omega$_m$ at gear re-engagement.

2) THE AVDIMC

In the AVDIMC, the controller switches its strategy according to the estimated system mode. Specifically, when the system is in the BM, two control strategies are employed to achieve both fast torque transmission and soft gear re-engagement: fast response control and impact mitigation (virtual damping) control. Fig. 15 illustrates the control regions in the $\hat{\theta}_b$–$\hat{\omega}_b$ plane. It defines Ω_f (fast response area) around the center of the backlash range and Ω_m (impact mitigation area) near the backlash boundary, which are used to select the control action during the BM. Fig. 15 illustrates the switching regions used in the AVDIMC control strategy, clarifying the transition between the fast-response region and the impact-mitigation region. Fig. 16 illustrates the physical behavior of the drivetrain under the AVDIMC. The left-hand illustration corresponds to the fast response control in Ω_f, while the right-hand illustration corresponds to the impact mitigation control in Ω_m at gear re-engagement. A block diagram of the overall control structure is shown in Fig. 17, which summarizes how the observer estimates are used for mode detection and for switching among the contact-mode control, fast response control, and impact mitigation control.

Control during the CMs ($\hat{\theta}_b = \pm\alpha$): This control is the same as the AVDC and is given by Eq. (19).

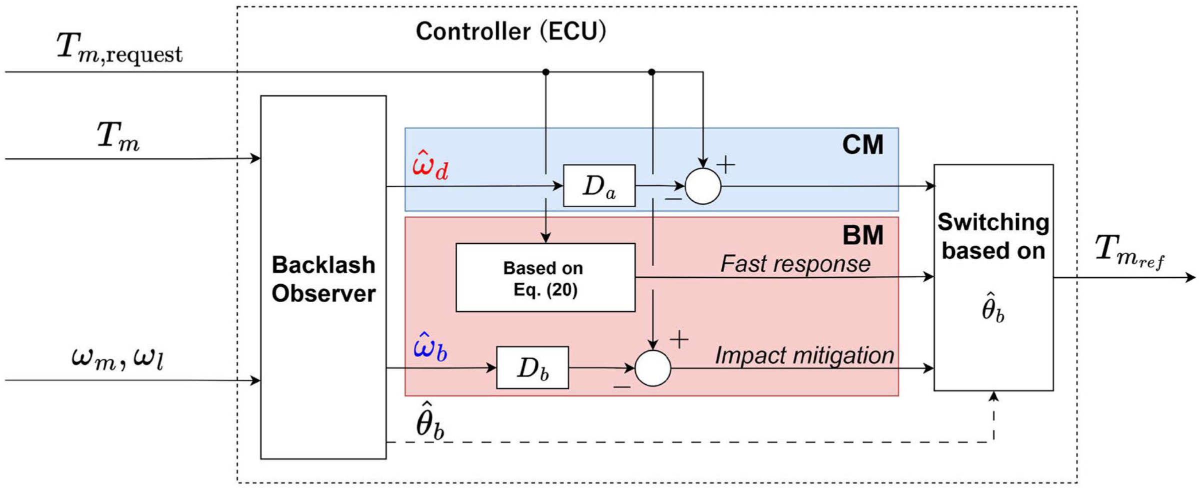

FIGURE 17. Block diagram of AVDIMC. The diagram shows the observer-based switching logic among contact-mode control, fast response control, and impact mitigation control during the backlash phase.

Fast response control in the BM ($(\hat{\theta}_b, \hat{\omega}_b) \in \Omega_f$): This control aims to achieve a fast exit from the BM and is executed when $(\hat{\theta}_b, \hat{\omega}_b) \in \Omega_f$. For a soft gear contact, which is achieved by impact mitigation control, the maximum torque of the motor is regulated by $T_{m,\text{fast}}$. The control input in this strategy, $T_{m,\text{ref}}$, is given as follows:

$$T_{m,\text{ref}} = \begin{cases} T_{m,\text{fast}}, & T_{m,\text{request}} > 0 \\ 0, & T_{m,\text{request}} = 0 \\ -T_{m,\text{fast}}, & T_{m,\text{request}} < 0 \end{cases} \tag{20}$$

Impact mitigation control in the BM ($(\hat{\theta}_b, \hat{\omega}_b) \in \Omega_m$): The impact mitigation area Ω_m is adjusted by $\Delta = \alpha/10$. This control is executed when $(\hat{\theta}_b, \hat{\omega}_b) \in \Omega_m$. Because the estimated backlash angular velocity $\hat{\omega}_b$ corresponds to the gear impact velocity, a negative feedback of $\hat{\omega}_b$ produces a pseudo-damping effect, which mitigates the impact caused by the gear collision. Consequently, the vibrations originating from the backlash can be suppressed. The control input in this strategy, $T_{m,\text{ref}}$, is expressed as follows:

$$T_{m,\text{ref}} = T_{m,\text{request}} - D_b\hat{\omega}_b,$$

where D_b is a feedback gain that should be, in general, larger than D_a.

The overall structure of the AVDIMC is to switch the three control strategies based on the estimate of the backlash angle $\hat{\theta}_b$ as given by Eq. (21).

$$T_{m,\text{ref}} = \begin{cases} T_{m,\text{request}} - D_a\hat{\omega}_d & \hat{\theta}_b = \pm\alpha \\ T_{m,\text{request}} - D_b\hat{\omega}_b & (\hat{\theta}_b, \hat{\omega}_b) \in \Omega_m \\ \begin{cases} T_{m,\text{fast}}, & T_{m,\text{request}} > 0 \\ 0, & T_{m,\text{request}} = 0 \\ -T_{m,\text{fast}}, & T_{m,\text{request}} < 0 \end{cases}, & (\hat{\theta}_b, \hat{\omega}_b) \in \Omega_f \end{cases} \tag{21}$$

The design parameters D_a, D_b, Δ, and $T_{m,\text{fast}}$ were set as shown in Table 7.

TABLE 7. Design parameters and values for AVDC and AVDIMC.

Parameter	Value
D_a	800
D_b	500
Δ	0.0057 rad
$T_{m,\text{fast}}$	1000 Nm

VI. PERFORMANCE EVALUATION WITH SIMULATOR

The following two subsections present the simulation results of the proposed vibration-suppression control methods. The simulation condition corresponds to a tip-in operation, in which the system transitions from regenerative braking to acceleration. Moreover, considering the experimental setup, the motor torque saturation is set to 20 Nm.

The evaluation points are described as follows. The torque command $T_{m,\text{request}}$, shown in Fig. 3(a), represents the tip-in operation. The control system is designed such that the motor output torque T_m tracks $T_{m,\text{request}}$ as accurately as possible while suppressing undesired vehicle vibrations caused by driveshaft torsion and gear backlash. Because the driving force applied to the vehicle is transmitted through the driveshaft, particular attention is paid to how quickly the driveshaft torsional torque T_s converges to $T_{m,\text{request}}$ without exhibiting undesired oscillations.

A. SIMULATION RESULTS OF THE AVDC

To evaluate the vibration-suppression performance of the AVDC, simulation results are presented together with the two conventional methods described in Section V. Figs. 18 and 19 compare the uncontrolled case and three control strategies. In these figures, the gray line indicates the uncontrolled response, the light blue line represents the response with the DCC, the purple line corresponds to the ω_m FB, and the red line represents the proposed AVDC.

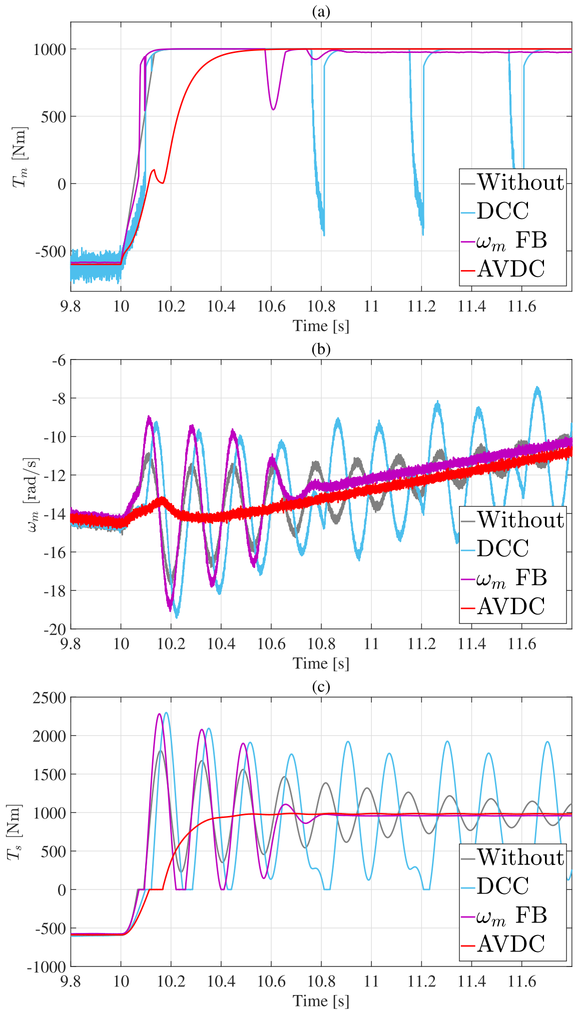

FIGURE 18. **Comparison of the proposed AVDC and conventional methods:(a) motor torque T_m, (b) motor-side angular velocity ω_m, and (c) driveshaft torsional torque T_s. Gray: without control; light blue: DCC; purple: ω_m FB; red: AVDC.**

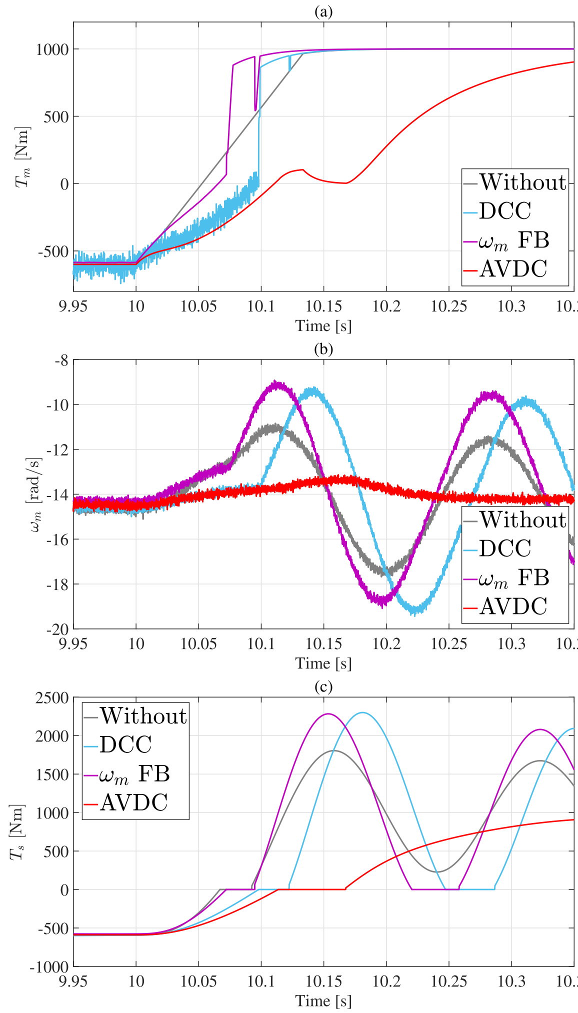

FIGURE 19. **Enlarged views around the step input corresponding to Fig. 18:(a) T_m, (b) ω_m, and (c) T_s.**

Fig. 18(a) shows the motor torque T_m. All controllers track the step-like command; however, differences appear immediately after torque reversal. Fig. 18(b) shows the motor-side angular velocity ω_m, where the conventional methods exhibit sustained oscillations caused by backlash-induced impacts, whereas the proposed AVDC significantly damps the oscillatory component. Fig. 18(c) shows the drive-shaft torsional torque T_s, which directly reflects driveline vibration. Here, large-amplitude oscillations persist for the conventional methods, while the proposed AVDC effectively suppresses backlash-induced torsional vibration.

To highlight the transient behavior, Fig. 19 provides enlarged views around the step input corresponding to Fig. 18(a)–(c). Fig. 19(a) magnifies the torque build-up in T_m, Fig. 19(b) emphasizes the transient oscillations in ω_m, and Fig. 19(c) clearly shows the difference in the convergence of the torsional torque T_s among the control methods. From Figs. 18 and 19, it is evident that the proposed AVDC achieves superior suppression of torsional vibrations compared with the conventional methods. In particular, significant differences are observed in the responses of T_s (Figs. 18(c) and 19(c)), where all controllers except for the AVDC are strongly affected by the backlash nonlinearity.

A closer examination reveals the following characteristics. For the ω_m FB, although the primary shaft resonance is attenuated, vibrations associated with gear impacts due to backlash persist until approximately 4.3 s (see Fig. 18(b)). For the DCC, the backlash nonlinearity induces low-frequency oscillations that do not appear in the linearized system without backlash. This behavior arises from the anti-resonance of the feedback system (approximately 1.7 Hz; see Fig. 11), which is excited by the backlash nonlinearity.

In contrast, the proposed AVDC suppresses torsional vibrations regardless of the presence of backlash. However, as shown in Figs. 18(c) and 19(c), the convergence of T_s is delayed, resulting in a longer idle duration. This behavior is explained by Eq. (19), which indicates that no control action is applied during the BM, where $\hat{\omega}_d = 0$. This limitation motivates the improved control strategy described in the next subsection.

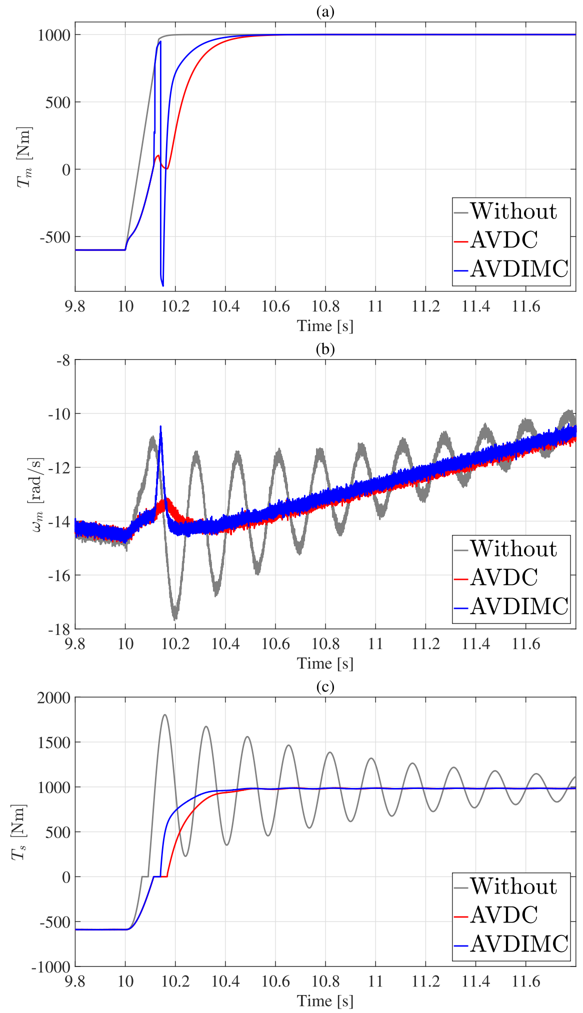

FIGURE 20. Comparison of the AVDIMC and the AVDC:(a) motor torque T_m, (b) motor-side angular velocity ω_m, and (c) driveshaft torsional torque T_s. Gray: without control; red: AVDC; blue: AVDIMC.

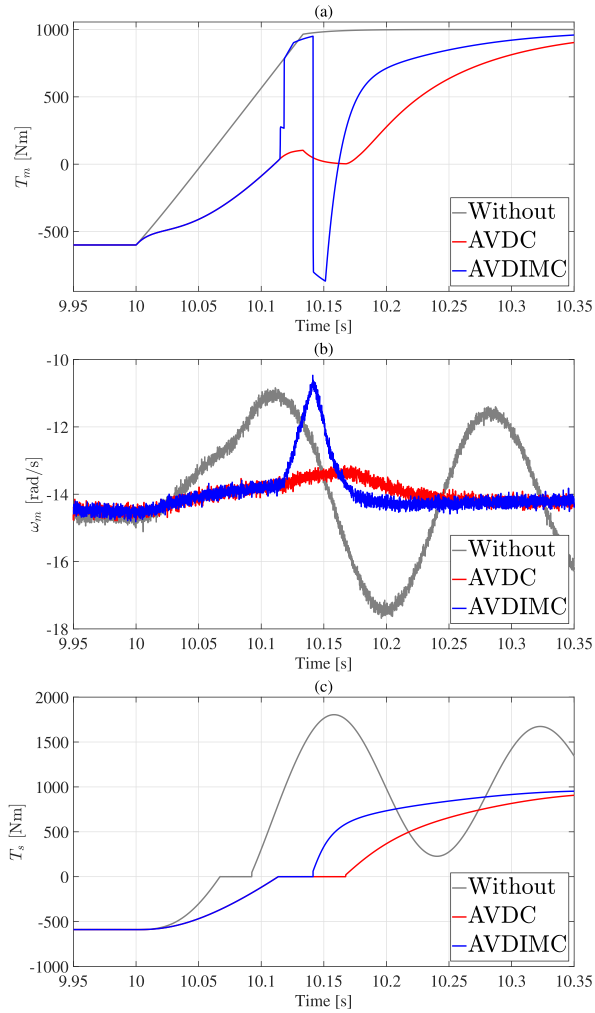

FIGURE 21. Enlarged views around the step input corresponding to Fig. 20:(a) T_m, (b) ω_m, and (c) T_s.

B. SIMULATION RESULTS OF THE AVDIMC

This subsection compares the performance of the AVDC and the AVDIMC based on the simulation responses shown in Fig. 20 and Fig. 21. The simulation conditions are identical to those used for the AVDC. Because the AVDC already outperforms the conventional methods, the comparison focuses on the additional benefits provided by the AVDIMC.

In Fig. 20 and Fig. 21, the gray line represents the uncontrolled response, whereas the red and blue lines correspond to the AVDC and the AVDIMC, respectively. Fig. 20(a) shows the motor torque T_m. Both control methods track the requested torque, while the AVDIMC exhibits a faster initial torque build-up. Fig. 20(b) shows the motor-side angular velocity ω_m, where both methods suppress sustained oscillations compared with the uncontrolled case. Fig. 20(c) shows the driveshaft torsional torque T_s, demonstrating that the AVDIMC maintains the vibration-suppression performance of the AVDC.

To highlight the transient behavior in detail, Fig. 21 provides enlarged views around the step input. Fig. 21(a) magnifies the torque response T_m, clearly showing the faster rise achieved by the AVDIMC. Fig. 21(b) shows the enlarged motor-side angular velocity ω_m, where transient oscillations are similarly attenuated by both methods. Fig. 21(c) shows the enlarged response of the driveshaft torsional torque T_s; the convergence time with the AVDIMC is reduced by approximately 60 % compared with that of the AVDC. Moreover, the idle duration is reduced from 0.0538 s (AVDC) to 0.0278 s (AVDIMC), corresponding to a 48.3 % reduction. This faster convergence indicates an earlier establishment of torque transmission and a shorter idle duration associated with the backlash mode, which is consistent with the fast-response control strategy defined in Eq. (20). Although a systematic sensitivity analysis with respect to drivetrain parameters such as inertia, stiffness, and damping is beyond the scope of this paper, the proposed observer and control framework remained stable in both simulation and vehicle experiments despite inevitable modeling uncertainties. A detailed parameter sensitivity analysis will be addressed in future work.

VII. REAL VEHICLE EVALUATION

This chapter presents vehicle experimental results to validate the effectiveness of the proposed control methods (the AVDC, the AVDIMC) under realistic operating conditions. The experiments were conducted using an electric vehicle (EV) modified for research purposes by AISIN Corporation, and the experiment planning and evaluation were performed jointly with their professional engineers.

A. EXPERIMENTAL SETUP AND TEST SCENARIO

Vehicle tests were carried out using a front-wheel-drive (FF) experimental EV. The vehicle was modified by AISIN Corporation for the purpose of research and development, and all tests were conducted in collaboration with AISIN on a dedicated proving ground.

As a test scenario, the vehicle was decelerated by regenerative braking from 45 km/h, and a tip-in maneuver was applied when the vehicle speed reached 35 km/h. This maneuver reproduces a transient condition in which the torque transmission direction is switched in an EV powertrain with gear backlash.

The input torque command was designed as a step-like input. For vehicle protection, the torque rate was limited to 12000 $\mathrm{N \cdot m/s}$, and the maximum output was also constrained. These constraints were introduced solely for safety reasons and are not considered to affect the relative comparison among the evaluated control methods in an essential manner.

To confirm repeatability under identical conditions, the same test scenario was repeated more than 10 times. Furthermore, the experiments were conducted not only with repeated runs on the same day but also across multiple days, and similar response tendencies were consistently observed. Due to space limitations, representative results are presented in this paper.

B. EXPERIMENTAL RESULTS

Fig. 22 shows representative vehicle responses during the tip-in maneuver. In each plot, the gray solid line represents the case without control, the red solid line represents the AVDC, and the blue solid line represents the AVDIMC. Fig. 22(a) shows the motor torque response, which confirms that the commanded torque is realized during the tip-in maneuver. Fig. 22(b) shows the driveshaft torsional torque, where backlash-induced oscillations are suppressed by both the AVDC and the AVDIMC compared with the uncontrolled case. Fig. 22(c) shows the longitudinal vehicle acceleration, indicating that the reduction in torsional oscillation is reflected in the vehicle-level response.

To highlight the transient behavior at the moment of the tip-in input, Fig. 23 provides magnified views corresponding to Fig. 22. Fig. 23(a) magnifies the torque build-up in the motor torque response immediately after tip-in. Fig. 23(b) magnifies the early transient of the driveshaft torsional torque, enabling direct comparison of oscillation amplitude and its decay among the three cases. This figure shows a substantial improvement of T_s convergence for the AVDIMC compared with the AVDC, which agrees with the simulation result in Fig. 21(c). Fig. 23(c) magnifies the longitudinal acceleration response and shows how the transient oscillations perceived at the vehicle level are mitigated by the proposed control methods.

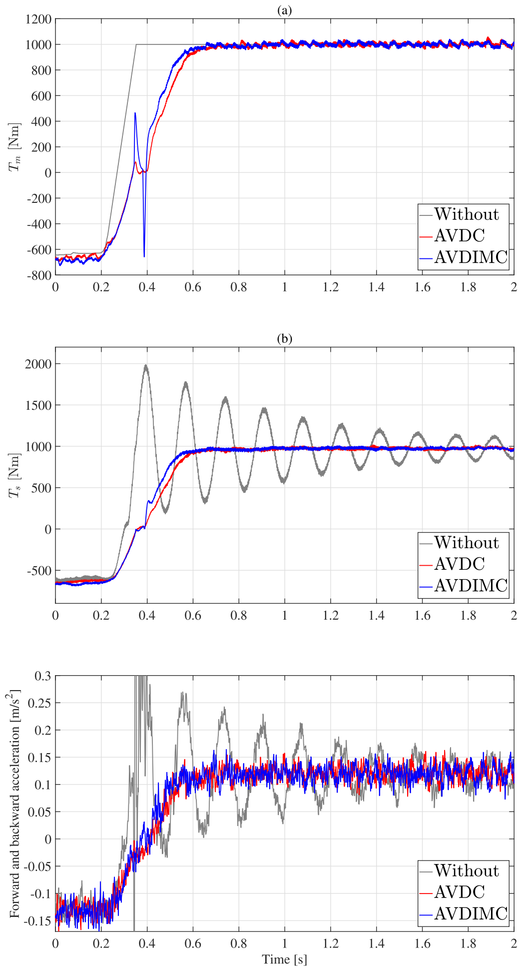

FIGURE 22. Measured vehicle responses during the tip-in maneuver. The plots show (a) motor torque, (b) driveshaft torsional torque, and (c) longitudinal acceleration to evaluate driveline vibration and its vehicle-level impact.

It should be noted that, due to the vehicle structure, an additional vibration component exists that is related to the mount structure fixing the motor unit to the vehicle body, which is distinct from the backlash-induced vibration. Its influence appears particularly in the initial part of the longitudinal acceleration waveform (Fig. 22(c)). However, this vibration component is observed commonly under all control conditions, and therefore it does not affect the comparative evaluation based on differences among the control methods.

C. QUANTITATIVE EVALUATION

This section quantitatively evaluates the vibration suppression performance of the proposed methods (the AVDC, the

TABLE 8. **Quantitative results in vehicle experiments (RMS evaluation window: t = 0–1.5 s after tip-in; raw signals). The metrics quantify vibration magnitude (peak and RMS) and decay characteristics (settling time) for the three cases.**

Metric	Without	The AVDC	The AVDIMC
Peak torsional torque [Nm]	≈ 2000	≈ 1000	≈ 1000
RMS of torsional torque [Nm]	1.01126×10^3	8.74727×10^2	8.86721×10^2
RMS of longitudinal acceleration [m/s^2]	0.13937	0.11208	0.11220
Settling time of T_s [s]	≈ 1.41	≈ 0.57	≈ 0.53

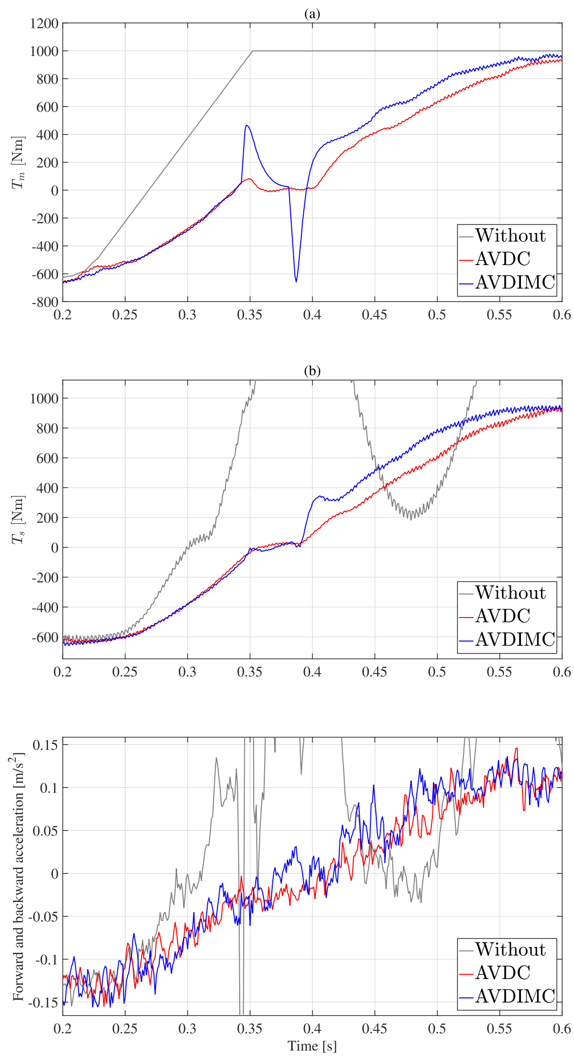

FIGURE 23. **Magnified views around the tip-in input corresponding to Fig. 22. The plots focus on the early transient in (a) motor torque, (b) driveshaft torsional torque, and (c) longitudinal acceleration for comparing decay and oscillation amplitude.**

AVDIMC) based on the vehicle test data. The performance indices are defined using the driveshaft torsional torque and the longitudinal vehicle acceleration.

The RMS values of the torsional torque and longitudinal acceleration were calculated from the raw (unfiltered) signals over a 1.5 s interval after the tip-in input, where the tip-in timing was set to $t = 0$. Since the torsional torque oscillation shows similar amplitudes in the positive and negative directions, an absolute-value-based evaluation was adopted for the peak torsional torque metric.

To evaluate the decay characteristics of the transient vibration, the settling time was introduced. The settling time was defined as the time required for the amplitude of the driveshaft torsional torque T_s to decay below 10 % of its peak value after the tip-in input. In experimental vehicle data, it is generally difficult to define a unique steady-state value due to residual fluctuations and measurement noise; therefore, a peak-based threshold was adopted to evaluate the transient decay.

Table 8 summarizes the quantitative results for the three cases (without control, the AVDC, and the AVDIMC). Compared with the case without control, both the AVDC and the AVDIMC reduce the RMS values of torsional torque and longitudinal acceleration. The settling time is shortened from approximately 1.41 s (without control) to approximately 0.57 s (the AVDC) and 0.53 s (the AVDIMC), indicating faster decay of transient oscillations.

D. DISCUSSION BASED ON VEHICLE EVALUATION

Compared with the case without control, both the AVDC and the AVDIMC reduce oscillations in the driveshaft torsional torque, confirming that the proposed methods function effectively on the actual vehicle. In particular, the peak torsional torque is reduced from approximately 2000 Nm (without control) to approximately 1000 Nm for both control methods.

The RMS torsional torque is reduced for both the AVDC and the AVDIMC compared with the case without control. Similarly, the RMS longitudinal acceleration is reduced for both the AVDC and the AVDIMC. These results are consistent with the qualitative observations in Figs. 22 and 23.

When comparing the AVDC and the AVDIMC, both methods show comparable reductions in RMS torsional torque and RMS longitudinal acceleration, whereas the settling time is slightly shorter for the AVDC under the present test condition. On the other hand, the AVDIMC is designed to emphasize torque transmission responsiveness, and it yields a faster initial build-up of the driveshaft torsional torque T_s in response to the tip-in input (see Fig. 23(b)). This indicates that the torque transmission is established earlier than with the AVDC.

VIII. CONCLUSION

In this study, observer-based vibration suppression methods for electric vehicle (EV) powertrains with gear backlash were proposed. A control-oriented two-inertia drivetrain model incorporating backlash nonlinearity was employed, and observer-based feedback controllers were designed to estimate unmeasured states and suppress driveline vibrations under practical implementation constraints.

Simulation results obtained using a high-fidelity exact backlash simulator confirmed that the proposed methods achieved effective vibration suppression through accurate observer estimation, outperforming conventional approaches even in the presence of communication delay, measurement noise, and sensor resolution limitations. The proposed methods, AVDC and AVDIMC, satisfied key requirements for practical vehicle applications, including robustness to rapid torque reversals and feasibility under a 1 ms sampling period.

Furthermore, the effectiveness of the proposed methods was experimentally validated using an actual electric vehicle. Vehicle experiments demonstrated that both AVDC and AVDIMC suppress transient driveline torsional vibrations and longitudinal acceleration oscillations during torque reversals. Quantitative evaluation showed that the peak driveshaft torsional torque was reduced from approximately 2000 Nm in the uncontrolled case to approximately 1000 Nm with the proposed controls. In addition, the RMS values of the torsional torque and longitudinal acceleration over a 1.5 s interval after the tip-in input were reduced by approximately 13% and 20%, respectively. The settling time of the transient torsional vibration was shortened from approximately 1.41 s to approximately 0.53–0.57 s, indicating faster decay of transient oscillations. Between the AVDC and the AVDIMC, the settling time of the AVDIMC is approximately 6 % shorter than that of the AVDC, as predicted by the simulations.

These results indicate that the proposed control-oriented formulation is sufficient to suppress driveline oscillations in the frequency range relevant to vehicle drivability and ride comfort, and that the proposed observer-based vibration suppression framework is feasible for real-time automotive embedded control systems.

The results confirm that the proposed observer can estimate backlash-related states with sufficient accuracy under practical sensing limitations and that the proposed control strategies effectively suppress torsional oscillations in EV powertrains.

Future work includes systematic robustness evaluation with respect to backlash variation and operating condition changes, and further investigation of parameter tuning to balance torque responsiveness and ride comfort.

APPENDIX A
APPROXIMATION OF BACKLASH ANGLE

We claim that the approximations $\theta_b \approx \theta_d$, $\dot{\theta}_b \approx \dot{\theta}_d$ are valid during the BM. These are derived as follows.

Recall that the shaft torsion angle θ_s is

$$\theta_s = \theta_m/i - \theta_3 = \theta_d - \theta_b$$

and the torsion torque is given by $T_s = k\theta_s + c\dot{\theta}_s$. During the BM, where its initial time instance is denoted by t_B, $T_s(t) = 0$ for $t \geqq t_B$. From this, we have

$$0 = k\theta_s(t) + c\dot{\theta}_s(t) \quad \text{or} \quad \dot{\theta}_s(t) = -\frac{k}{c}\theta_s(t). \tag{22}$$

This is a first-order linear differential equation. Letting $\sigma = k/c$, where σ has the dimension of $[\mathrm{T}^{-1}]$, the solution is given by

$$\theta_s(t) = \theta_s(t_B)e^{-\sigma(t-t_B)} \tag{23}$$

for $t \geqq t_B$. From Eq. (23), for t such that $1 \ll \sigma(t - t_B)$, we have $\theta_s(t) \approx 0$, namely,

$$\theta_d(t) \approx \theta_b(t).$$

The validity of the approximation $\dot{\theta}_b \approx \dot{\theta}_d$ is shown in the same way by taking the derivative of (22).

In our system as well as in real vehicle drive shafts, $\sigma \approx 1000$ almost always holds, making it possible to apply these approximations to EV powertrains. For instance, $e^{-\sigma(t-t_B)} = 1/5,\ 1/10$, and $1/20$ for $t - t_B = 1.61$ ms, 2.30 ms, and 3.00 ms, respectively, which indicates that the approximation becomes valid almost immediately after the BM starts. As is already discussed in Section IV, these approximations help eliminate the backlash variable from the observer equations and thereby simplify the observer structure.

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MANATO KAWASAKI received the B.E. degree in mechanical and electronic control engineering from the Faculty of Science and Engineering, Nanzan University, Nagoya, Japan, in March 2024, where he is currently pursuing the master's degree with the Department of Mechanical and Electronic Control Engineering, Graduate School of Science and Technology. His research interests include powertrain and vehicle systems, vibration control, electric vehicles (EVs), mechanical backlash, and observer-based control.

NOBORU SAKAMOTO (Member, IEEE) received the B.Sc. degree in mathematics from Hokkaido University, in 1991, and the M.Sc. and Ph.D. degrees in aerospace engineering from Nagoya University, Nagoya, Japan, in 1993 and 1996, respectively. From 1996 to 2015, he held positions with the Graduate School of Engineering, Nagoya University. Currently, he is a Professor with the Science and Engineering, Nanzan University. He has held visiting research positions with the University of Groningen, The Netherlands, from 2005 to 2006, and the University of Deusto, Spain, from 2018 to 2019. His research interests include nonlinear control theory, dynamical system theory, and control applications for mechatronics and aerospace engineering. He received the SICE Best Paper Prizes, in 1997, 2006, 2008 and 2011; and the Kimura Prize, in 2016.

AKIRA NAKASHIMA (Member, IEEE) received the Ph.D. degree in engineering from Nagoya University, Nagoya, Japan, in 2005. He was an Assistant Professor with the Graduate School of Engineering, Nagoya University, from 2005 to 2015. He has been a Professor with the Faculty of Science and Engineering, Nanzan University, Japan, since 2020. His research interests include multi-fingered robot hands, dynamic manipulation, nonholonomic systems, model predictive control, and autonomous driving.

YASUHIKO KOBAYASHI received the B.E. degree in mechanical and aerospace engineering and the M.E. degree in mechanical system engineering from Nagoya University, Japan, in 1998 and 2000, respectively. In 2000, he joined Aisin AW Company Ltd. (currently Aisin Corporation), and has been engaged in advanced development in the field of electrification.

YUTARO TANAKA received the B.E. degree in mechanical and aerospace engineering from Nagoya University, Japan, in 2013. In 2013, he joined Aisin Seiki Company Ltd. (currently Aisin Corporation), and has been engaged in advanced development in the field of electrification.